Create mesh based on ① $\boxed{\text{box}}$ as initail_bounary_box

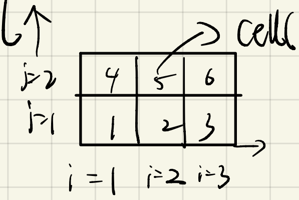
of 4 vertices
cooridnate (x_0, x_1, y_0, y_1)
1 2 3 4

number of cell ② n_x ③ n_y ∼ only make mesh

Once we have n_x, n_y, we know we have $n_x n_y$ cells $(n_x+1)(n_y+1)$ Vertex

e.g. $n_x=3$

Label the cell

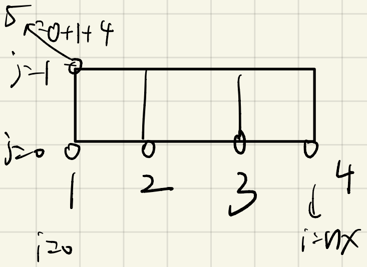


$\text{cell}(i,j,n_x) = i + n_x(j-1)$ ∼ Cell_index

start with (1,1).

$i=1,\cdots,n_x$
$j=1,\cdots,n_y$

Label init vertex



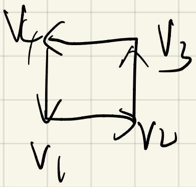
$\text{vertex\_idx}(i,j,n_x)$ ∼ vertex_index
$= i+1+j(n_x+1)$

$i=0,\cdots,n_x$ $j=0,\cdots,n_y$

So we save all cell with corresponding vertex index

in ④ Cells_vertex_index_mat

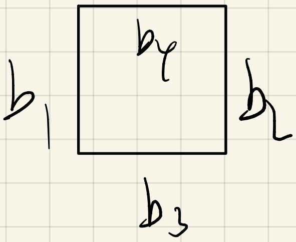
$$\begin{bmatrix} \text{cell}_1 & & \\ V_1 & & \\ \vdots & \cdots & \\ V_4 & & \end{bmatrix}$$



Then we can lable each faces to the list

in ⑤ faces_list [face1, ⋯ face n]

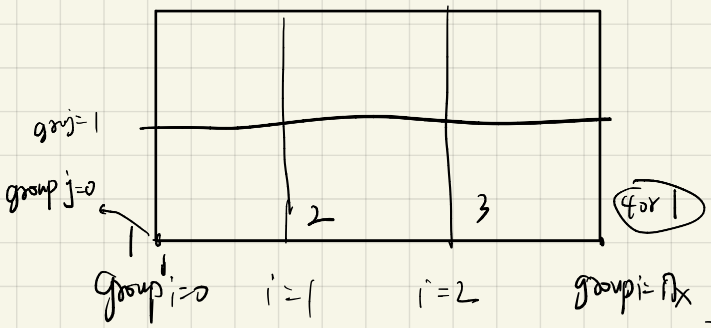
And boundary types $[b_1, b_2, b_3, b_4]$



⑥ boundaries

for period boundary condition, we need maps

each vertexs to it's group index ⑦ vertex2group_index $\begin{array}{c} V1 \quad\quad VN \\ [g1 --- g_i] \\ \quad\quad V \to g_i \end{array}$



group i = n_x if not periodic on Left & right i=3

if periodic → group n_x mod n_x = 0 i=0

$$\text{group index}: (i, j, n_x) = (\text{group } i)+1 + \text{group } j\,(\overset{n_x \text{ or } n_x+1}{\max(\text{group } i)} - 1)$$

Create index for each Vertexs as columns of X
DG.state

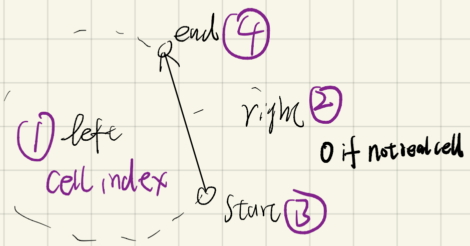
$$\begin{array}{ccc} V_1, & --- & , V_N \\ \Big[\begin{matrix} x_i \\ y_i \end{matrix} & & \Big] \end{array}$$

only coordinate change with time

index fixed

vertices_coordinate_mat

Create each face



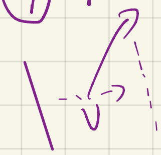
0 if not real cell

⑤ Whether contact (not flux)

⑥ boundary type

⑦ period

shift vector



iterate each cell for it's right face, Top face

Then see boundary to decide include Left, Bottom,